

Cache Documentation

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Overview

The Proxy Service implements a comprehensive caching system to reduce bandwidth usage and improve response times.

Cache Types

1. Squid HTTP Cache

The primary caching mechanism for HTTP/HTTPS content.

Configuration:

```
CACHE_DIR=/path/to/cache/squid
CACHE_MAX_SIZE_GB=50
CACHE_MAX_OBJECT_SIZE_MB=100
CACHE_MIN_OBJECT_SIZE_KB=1
CACHE_MEMORY_SIZE_MB=512
```

Storage Format: AUFS (Asynchronous Unlinkable File System)

- Level 1: 16 directories
- Level 2: 256 subdirectories per level 1

2. Streaming Cache

Specialized cache for video/audio streaming content.

Configuration:

```
STREAMING_CACHE_ENABLED=true
STREAMING_CACHE_MAX_SIZE_GB=20
STREAMING_DOMAINS=youtube.com,googlevideo.com
STREAMING_CHUNK_SIZE_KB=256
```

Features:

- Chunk-based caching for partial content
- Range request support
- Separate storage from HTTP cache

Cache Storage

Directory Structure

```
CACHE_DIR/
├── squid/
│   ├── 00/
│   │   ├── 00/
│   │   │   ├── 00000001 # Cached object
│   │   │   ├── 00000002
│   │   │   └── ...
│   │   ├── 01/
│   │   └── ... (256 dirs)
│   ├── 01/
│   └── ... (16 dirs)
└── streaming/
    ├── video/
    │   ├── youtube_abc123_chunk1
    │   └── ...
    ├── audio/
    └── manifest/
```

Object Naming

Squid stores objects using hexadecimal naming:

- 8-character filename
- Based on hash of URL and headers
- Spread across directory levels

Cache Policies

Refresh Patterns

Pattern	Min	Percent	Max	Options
^ftp:	1440	20%	10080	-
^gopher:	1440	0%	1440	-
cgi-bin	0	0%	0	-
default (.)	0	20%	4320	-

Explanation:

- **Min:** Minutes before checking freshness
- **Percent:** Percentage of age for stale check
- **Max:** Maximum age without revalidation

Cacheability Rules

Cacheable by default:

- GET responses
- 200, 203, 300, 301, 302, 307, 410, 404 responses
- Responses with Cache-Control: public

Not cacheable:

- POST, PUT, DELETE requests
- Responses with Cache-Control: no-store
- Responses with Authorization header (unless public)
- Responses with Set-Cookie header

Cache Management

Automatic Invalidation

Run by cache-invalidator container:

```
# Interval  
CACHE_INVALIDATE_INTERVAL=24 # hours
```

```
# Max age  
CACHE_MAX_AGE_DAYS=30
```

```
# Max size  
CACHE_MAX_SIZE_GB=50
```

Process:

1. Remove files older than CACHE_MAX_AGE_DAYS
2. Remove empty directories
3. Trim to CACHE_MAX_SIZE_GB if over limit

Manual Commands

```
# View statistics  
./cachectl stats
```

```
# Clear all cache  
./cachectl clear
```

```
# Run invalidation
./cachectl invalidate
```

```
# Trim to size
./cachectl trim 30
```

Invalidation Triggers

Trigger	Action
Age exceeded	Delete old files
Size exceeded	Delete oldest files
VPN reconnect	Optionally clear cache
Manual command	User-initiated cleanup

Cache Hit/Miss Flow

Cache Hit

```
Request → Squid → Cache Lookup → [HIT] → Response
                                   |
                                   └─ Return cached content immediately
```

Cache Miss

```
Request → Squid → Cache Lookup → [MISS] → Origin → Response
                                   |
                                   └─ Forward request, store response
```

Stale Content

```
Request → Squid → Cache Lookup → [STALE] → Revalidate → Response
                                   |                               |
                                   |                               └─ Check if modified
                                   |
                                   └─ If 304: Return cached
                                       If 200: Update and return
```

Performance Tuning

Memory Cache

In-memory cache for hot objects:

```
CACHE_MEMORY_SIZE_MB=512
```

Recommendations:

- 10-20% of total cache size
- Increase for high-traffic scenarios
- Monitor hit ratio to optimize

Object Size Limits

```
# Maximum object size to cache  
CACHE_MAX_OBJECT_SIZE_MB=100
```

```
# Minimum object size to cache  
CACHE_MIN_OBJECT_SIZE_KB=1
```

Recommendations:

- Set max to prevent caching large files
- Set min to avoid caching tiny responses
- Adjust based on available disk space

Disk I/O

For high-traffic scenarios:

- Use SSD for cache directory
- Consider RAID for redundancy
- Monitor disk I/O metrics

Monitoring

Cache Statistics

```
./cachectl stats
```

Output includes:

- Total size
- File count
- Usage percentage
- Per-cache breakdown

Squid Cache Manager

Access Squid's built-in cache manager:

```
# Via cachemgr.cgi or command line  
squidclient mgr:info
```

```
squidclient mgr:mem  
squidclient mgr:storedir
```

Key Metrics

Metric	Description	Target
Hit Ratio	% of requests served from cache	> 30%
Byte Hit Ratio	% of bytes served from cache	> 20%
Request Rate	Requests per second	Varies
Response Time	Average response time	< 100ms

Streaming Cache Details

Supported Domains

Configure domains for streaming cache:

`STREAMING_DOMAINS=youtube.com,googlevideo.com,ytimg.com,vimeo.com`

Chunk Management

Streaming content is cached in chunks:

- Default chunk size: 256KB
- Range requests handled separately
- Partial content supported

Video Streaming

For YouTube and similar services:

1. Manifest files cached separately
2. Video segments cached by range
3. Adaptive streaming supported

Troubleshooting

Cache Not Growing

1. Check disk space
2. Verify permissions
3. Check Squid logs

High Miss Ratio

1. Review cacheability rules
2. Check refresh patterns
3. Increase cache size

Corrupted Cache

Stop service

`./stop`

Clear cache

`./cachectl clear -f`

Rebuild cache structure

`./init`

Restart

`./start`

Cache Directory Full

Check size

`./cachectl stats`

Trim to size

`./cachectl trim 30`

Or increase limit in .env

`CACHE_MAX_SIZE_GB=100`